Introduction to VLSI Design

Lecture 7 Interconnect

Zhuo Feng

Outline

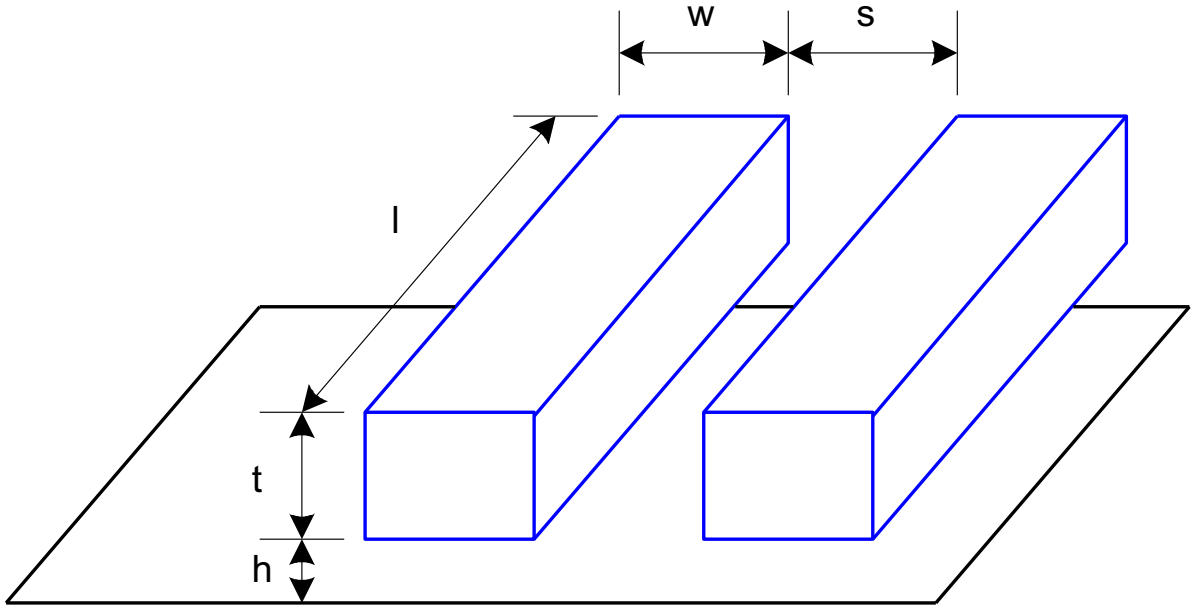
- Introduction
- Wire Resistance
- Wire Capacitance
- Wire RC Delay
- Crosstalk
- Wire Engineering
- Repeaters

Introduction

- **Chips are mostly made of wires called *interconnect***
  - In stick diagram, wires set size
  - Transistors are little things under the wires
  - Many layers of wires
- **Wires are as important as transistors**
  - Speed
  - Power
  - Noise
- **Alternating layers run orthogonally**

Wire Geometry

- **Pitch = w + s**
- **Aspect ratio: AR = t/w**
  - Old processes had AR << 1
  - Modern processes have AR ≈ 2
    - Pack in many skinny wires

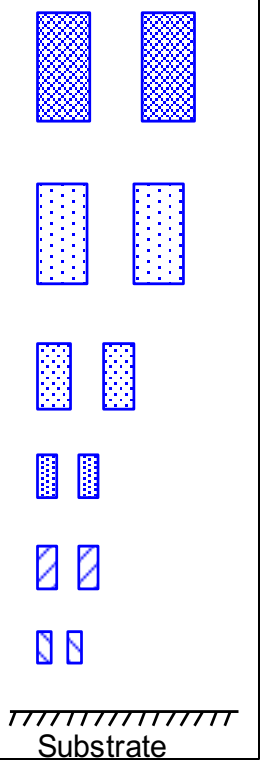


Layer Stack

- **AMI 0.6 μm process has 3 metal layers**
- **Modern processes use 6-10+ metal layers**
- **Example:**

  Intel 180 nm process
- **M1: thin, narrow (< 3λ)**
  - ► High density cells
- **M2-M4: thicker**
  - ► For longer wires
- **M5-M6: thickest**
  - ► For V_{DD}, GND, clk

Layer	T (nm)	W (nm)	S (nm)	AR
6	1720	860	860	2.0
	1000			
5	1600	800	800	2.0
	1000			
4	1080	540	540	2.0
	700			
3	700	320	320	2.2
	700			
2	700	320	320	2.2
	700			
1	480	250	250	1.9
	800			



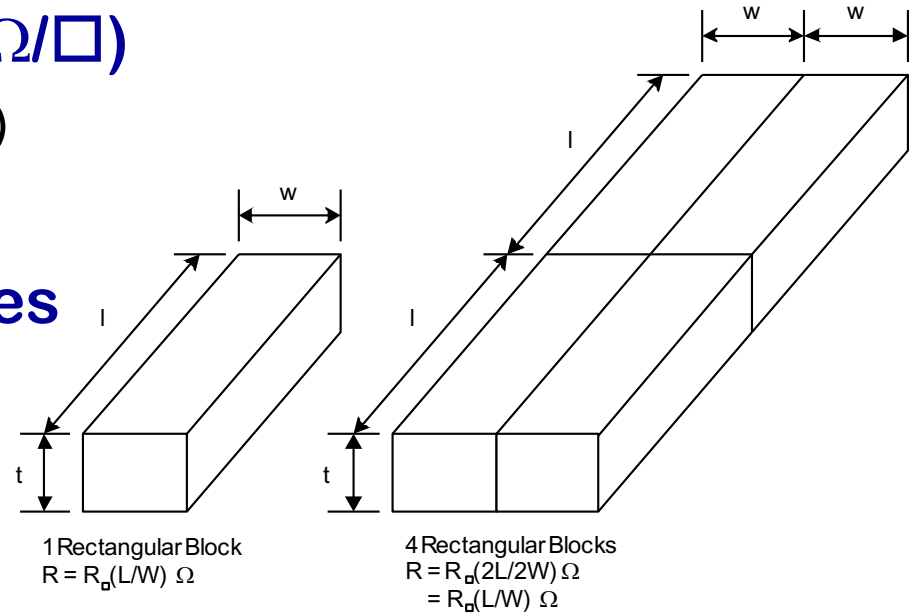
Wire Resistance

- **ρ = *resistivity* (Ω*m)**

$$R = \frac{\rho}{t}\frac{l}{w} = R_{\square}\frac{l}{w}$$

- **$R_{\square}$ = *sheet resistance* ($\Omega/\square$)**
  - ► $\square$ is a dimensionless unit(!)

- **Count number of squares**
  - ► $R = R_{\square}$ * (# of squares)



Choice of Metals

- **Until 180 nm generation, most wires were aluminum**
- **Modern processes often use copper**
  - Cu atoms diffuse into silicon and damage FETs
  - Must be surrounded by a diffusion barrier

Metal	**Bulk resistivity ($\mu\Omega$*cm)**
Silver (Ag)	1.6
Copper (Cu)	1.7
Gold (Au)	2.2
Aluminum (Al)	2.8
Tungsten (W)	5.3
Molybdenum (Mo)	5.3

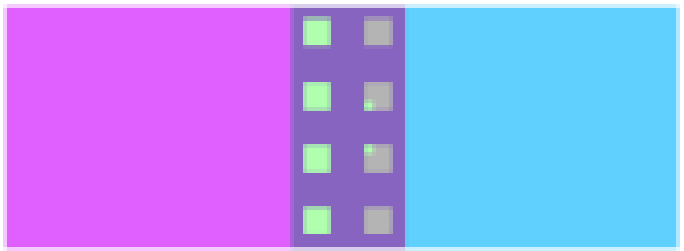
Sheet Resistance

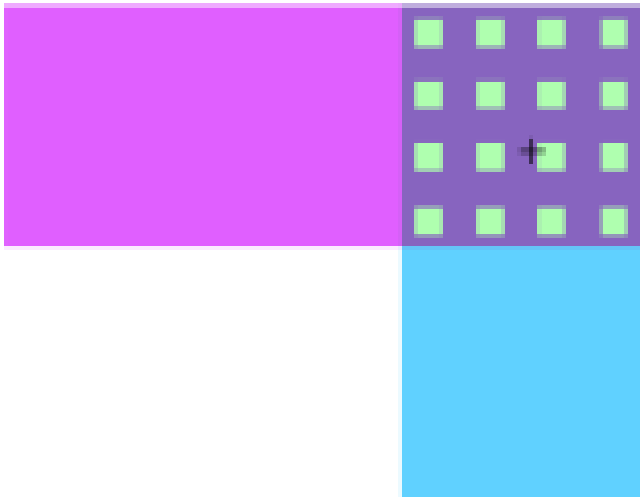
- **Typical sheet resistances in 180 nm process**

Layer	Sheet Resistance (Ω/□)
Diffusion (silicided)	3-10
Diffusion (no silicide)	50-200
Polysilicon (silicided)	3-10
Polysilicon (no silicide)	50-400
Metal1	0.08
Metal2	0.05
Metal3	0.05
Metal4	0.03
Metal5	0.02
Metal6	0.02

Contacts Resistance

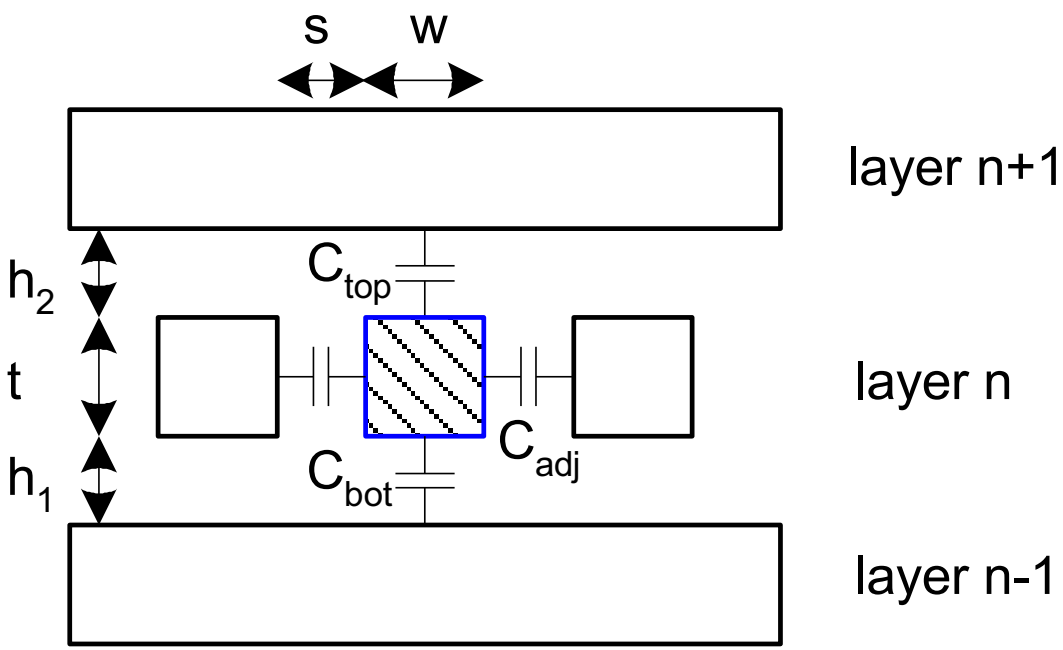
- **Contacts and vias also have 2-20 Ω**
- **Use many contacts for lower R**
  - Many small contacts for current crowding around periphery





Wire Capacitance

- **Wire has capacitance per unit length**
  - To neighbors
  - To layers above and below
- **$C_{total} = C_{top} + C_{bot} + 2C_{adj}$**

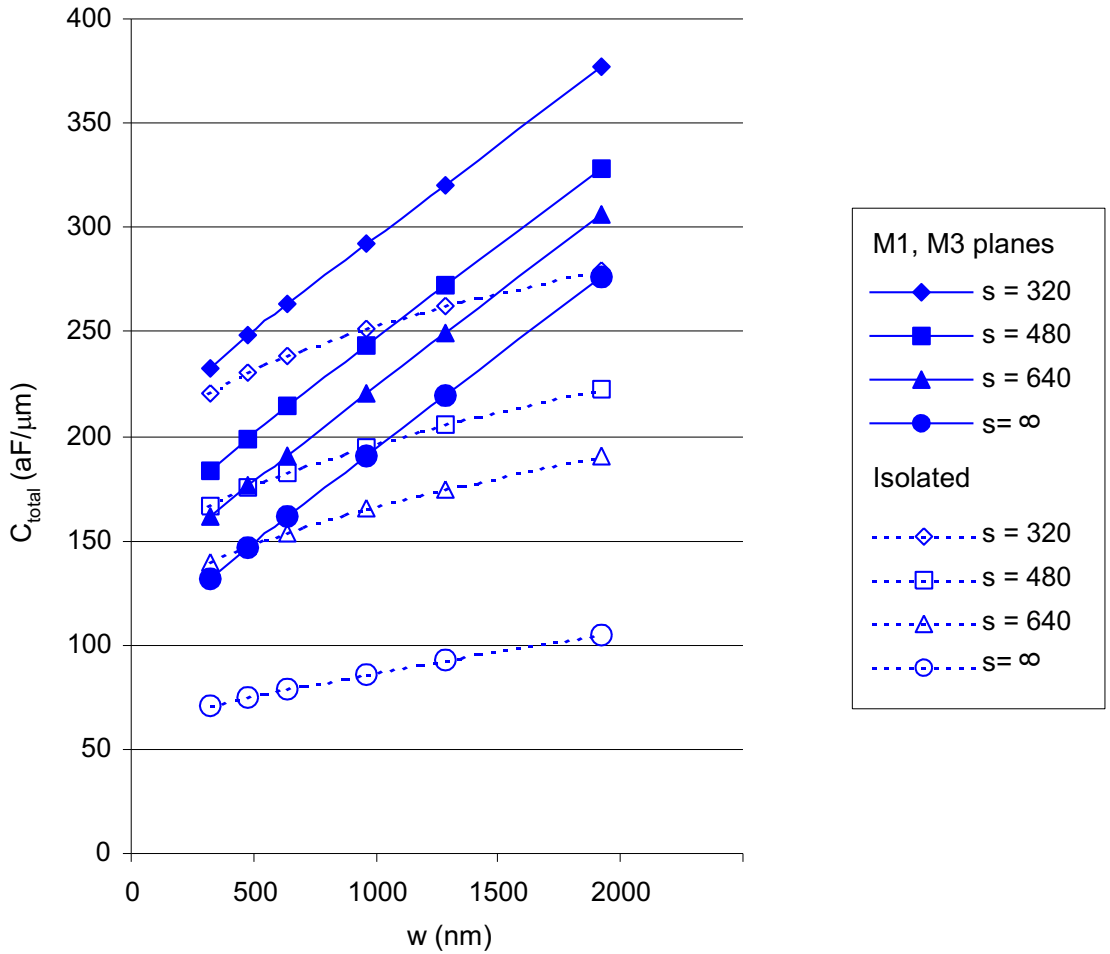


Capacitance Trends

- **Parallel plate equation: $C = \varepsilon A/d$**
  - Wires are not parallel plates, but obey trends
  - Increasing area (W, t) increases capacitance
  - Increasing distance (s, h) decreases capacitance
- **Dielectric constant**
  - $\varepsilon = k\varepsilon_0$
- **$\varepsilon_0 = 8.85 \times 10^{-14}$ F/cm**
- **k = 3.9 for SiO_2**
- **Processes are starting to use low-k dielectrics**
  - $k \approx 3$ (or less) as dielectrics use air pockets

M2 Capacitance Data

- **Typical wires have ~ 0.2 fF/μm**
  - Compare to 2 fF/μm for gate capacitance

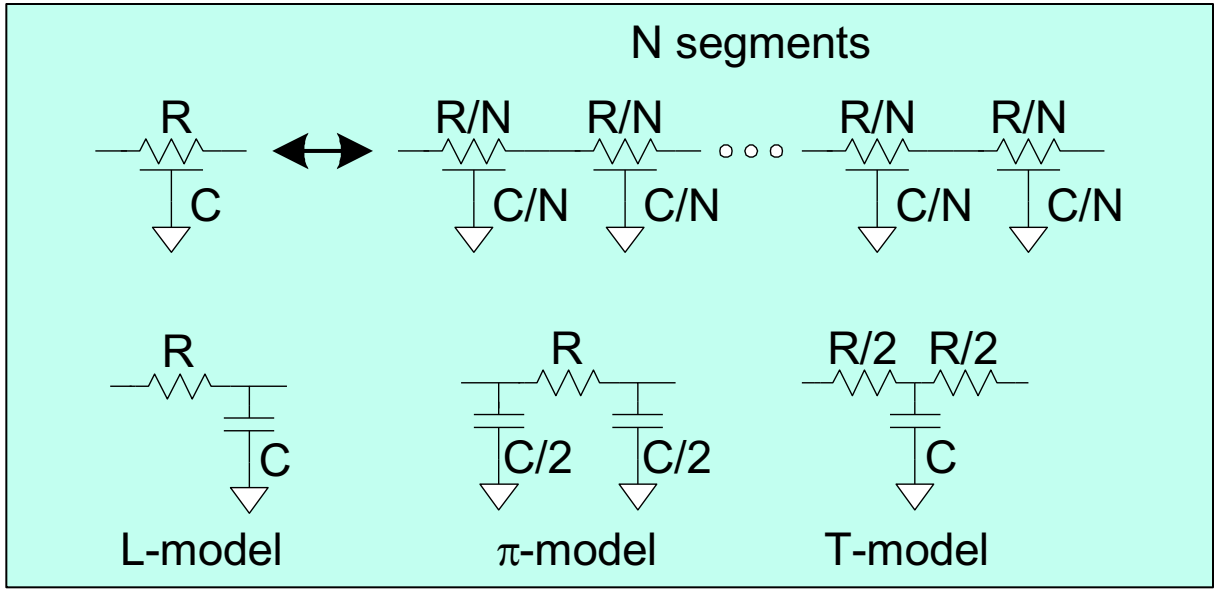


Diffusion & Polysilicon

- **Diffusion capacitance is very high (about 2 fF/μm)**
  - Comparable to gate capacitance
  - Diffusion also has high resistance
  - Avoid using diffusion *runners* for wires!
- **Polysilicon has lower C but high R**
  - Use for transistor gates
  - Occasionally for very short wires between gates

Lumped Element Models

- **Wires are a distributed system**
  - Approximate with lumped element models



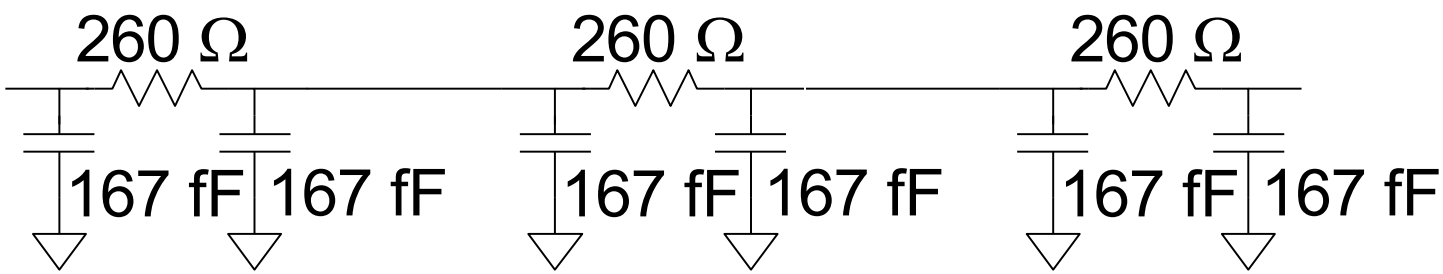
- **3-segment π-model is accurate to 3% in simulation**
- **L-model needs 100 segments for same accuracy!**
- **Use single segment π-model for Elmore delay**

Example

- **Metal2 wire in 180 nm process**
  - 5 mm long
  - 0.32 μm wide
- **Construct a 3-segment π-model**
  - $R_{\square}$ =
  - $C_{permicron}$ =

Example

- **Metal2 wire in 180 nm process**
  - 5 mm long
  - 0.32 μm wide
- **Construct a 3-segment π-model**
  - $R_{\square}$ = 0.05 Ω/□ => R = 781 Ω
  - $C_{permicron}$ = 0.2 fF/μm => C = 1 pF

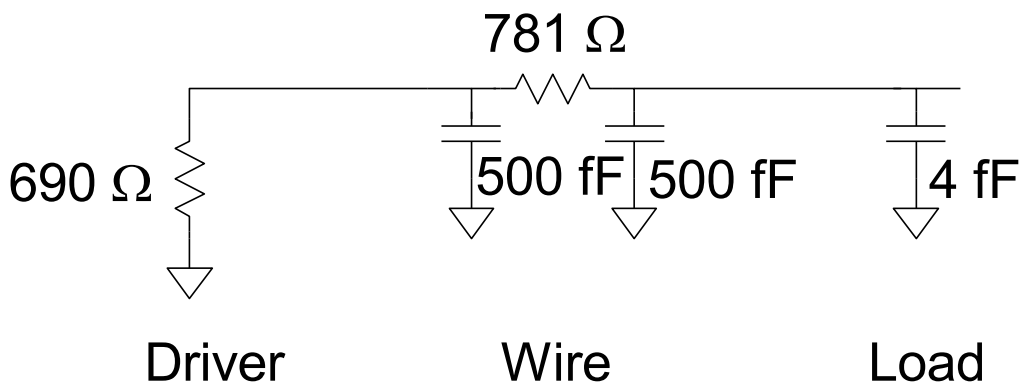


Wire RC Delay

- **Estimate the delay of a 10x inverter driving a 2x inverter at the end of the 5mm wire from the previous example.**
  - R = 2.5 kΩ*μm for gates
  - Unit inverter: 0.36 μm nMOS, 0.72 μm pMOS
  - t_{pd} =

Wire RC Delay

- **Estimate the delay of a 10x inverter driving a 2x inverter at the end of the 5mm wire from the previous example.**
  - R = 2.5 kΩ*μm for gates
  - Unit inverter: 0.36 μm nMOS, 0.72 μm pMOS



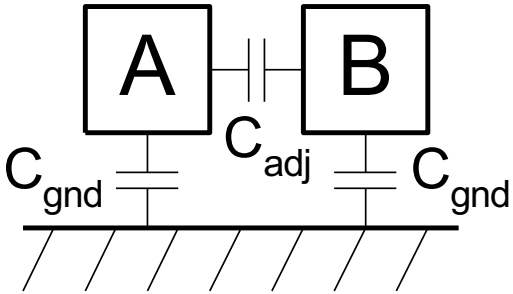
  - t_{pd} = 1.1 ns

Crosstalk

- **A capacitor does not like to change its voltage instantaneously.**
- **A wire has high capacitance to its neighbor.**
  - When the neighbor switches from 1-> 0 or 0->1, the wire tends to switch too.
  - Called capacitive *coupling* or *crosstalk*.
- **Crosstalk effects**
  - Noise on nonswitching wires
  - Increased delay on switching wires

Crosstalk Delay

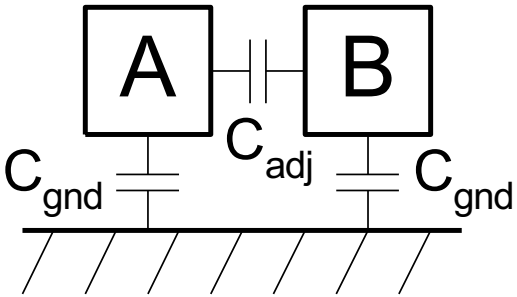
- **Assume layers above and below on average are quiet**
  - Second terminal of capacitor can be ignored
  - Model as $C_{gnd} = C_{top} + C_{bot}$
- **Effective C_{adj} depends on behavior of neighbors**
  - *Miller effect*



B	ΔV	$C_{eff(A)}$	MCF
Constant			
Switching with A			
Switching opposite A			

Crosstalk Delay

- **Assume layers above and below on average are quiet**
  - Second terminal of capacitor can be ignored
  - Model as $C_{gnd} = C_{top} + C_{bot}$
- **Effective C_{adj} depends on behavior of neighbors**
  - *Miller effect*

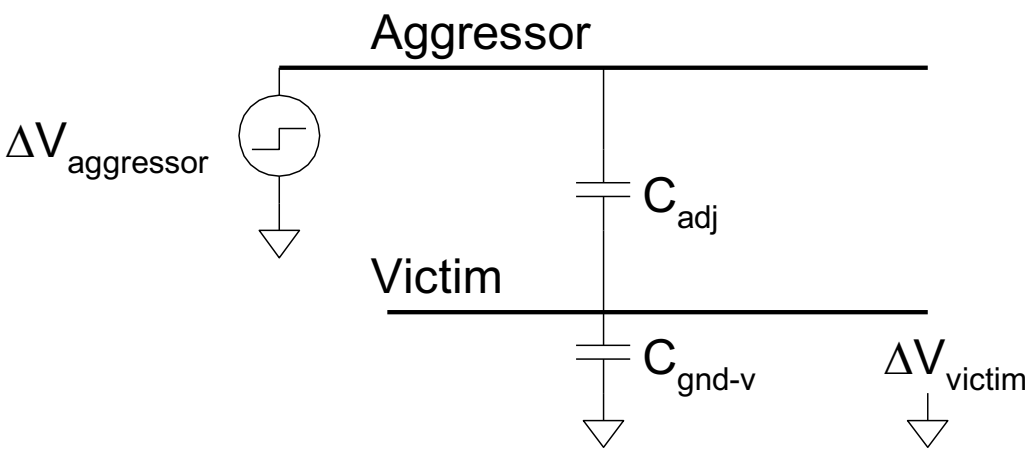


B	ΔV	$C_{eff(A)}$	MCF
Constant	V_{DD}	$C_{gnd} + C_{adj}$	1
Switching with A	0	C_{gnd}	0
Switching opposite A	$2V_{DD}$	$C_{gnd} + 2\ C_{adj}$	2

Crosstalk Noise

- **Crosstalk causes noise on nonswitching wires**
- **If victim is floating:**
  - model as capacitive voltage divider

$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd-v} + C_{adj}} \Delta V_{aggressor}$$

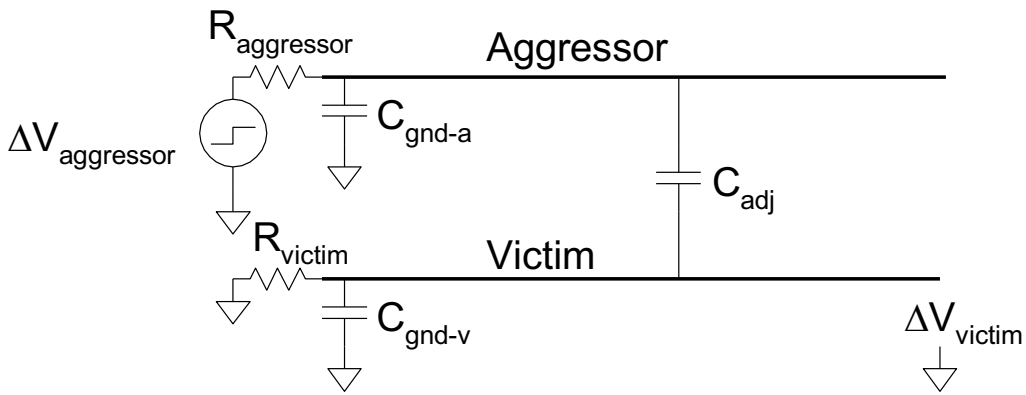


Driven Victims

- **Usually victim is driven by a gate that fights noise**
  - Noise depends on relative resistances
  - Victim driver is in linear region, agg. in saturation
  - If sizes are same, $R_{aggressor}$ = 2-4 x R_{victim}

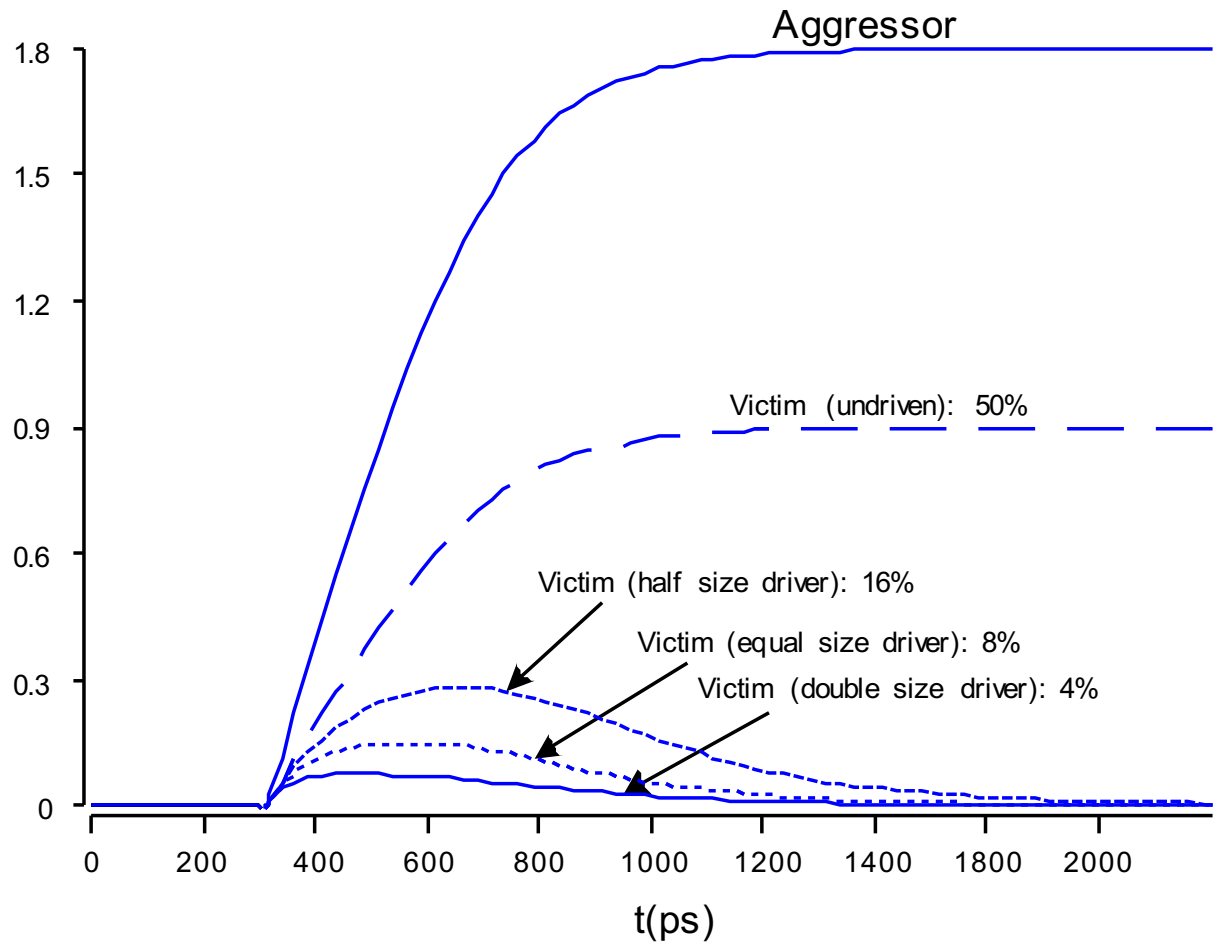
$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd-v} + C_{adj}} \frac{1}{1+k} \Delta V_{aggressor}$$

$$k = \frac{\tau_{aggressor}}{\tau_{victim}} = \frac{R_{aggressor}\left(C_{gnd-a} + C_{adj}\right)}{R_{victim}\left(C_{gnd-v} + C_{adj}\right)}$$



Coupling Waveforms

- **Simulated coupling for $C_{adj} = C_{victim}$**

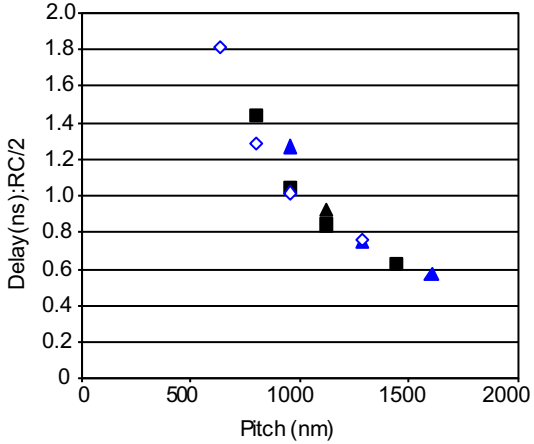


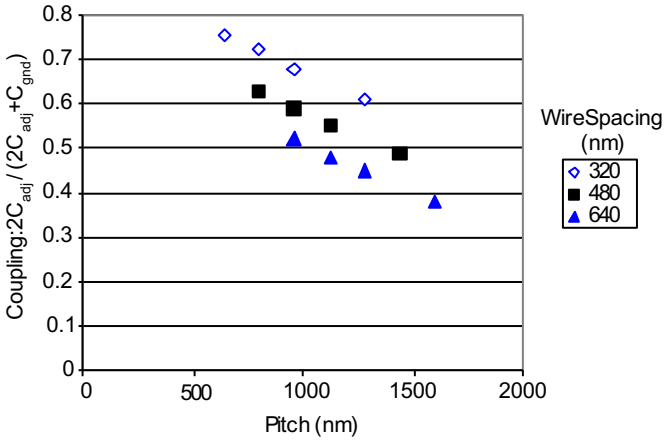
Noise Implications

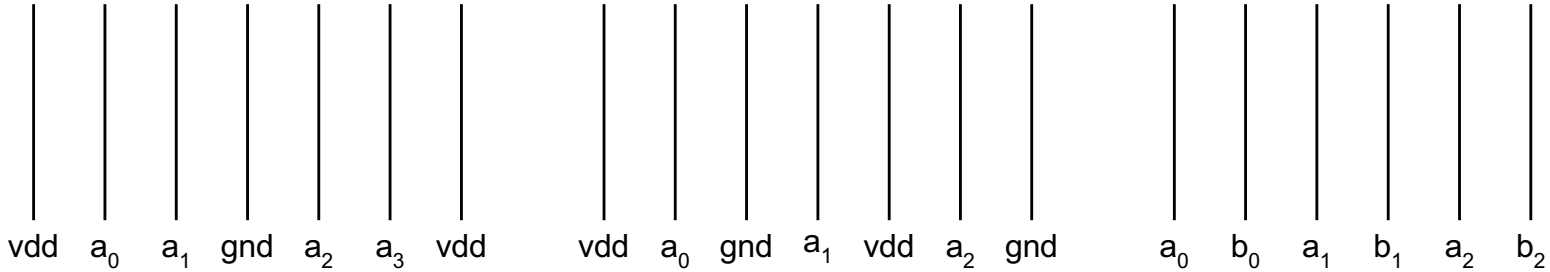
- ***So what*** **if we have noise?**
- **If the noise is less than the noise margin, nothing happens**
- **Static CMOS logic will eventually settle to correct output even if disturbed by large noise spikes**
  - But glitches cause extra delay
  - Also cause extra power from false transitions
- **Dynamic logic never recovers from glitches**
- **Memories and other sensitive circuits also can produce the wrong answer**

Wire Engineering

- **Goal: achieve delay, area, power goals with acceptable noise**
- **Degrees of freedom:**
  - Width
  - Spacing
  - Layer
  - Shielding

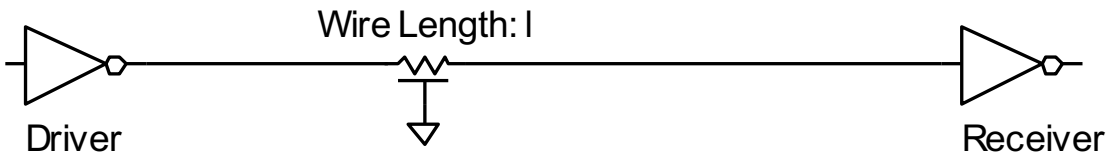


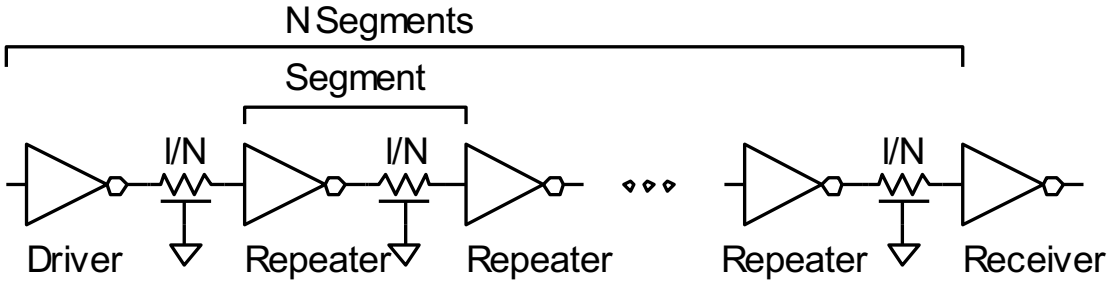




Repeaters (buffers)

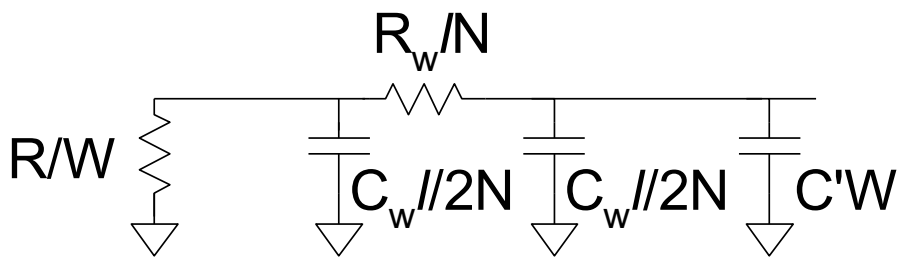
- **R and C are proportional to *l***
- **RC delay is proportional to l^2**
  - Unacceptably great for long wires
- **Break long wires into N shorter segments**
  - Drive each one with an inverter or buffer





Repeater Design

- **How many repeaters should we use?**
- **How large should each one be?**
- **Equivalent Circuit**
  - Wire length *l*
    - Wire Capacitance C_w*l, Resistance R_w*l
  - Inverter width W (nMOS = W, pMOS = 2W)
    - Gate Capacitance C'*W, Resistance R/W



Repeater Results

- **Write equation for Elmore Delay**
  - Differentiate with respect to W and N
  - Set equal to 0, solve

$$\frac{l}{N} = \sqrt{\frac{2RC'}{R_w C_w}}$$

$$\frac{t_{pd}}{l} = \left(2 + \sqrt{2}\right)\sqrt{RC'R_w C_w}$$

$$W = \sqrt{\frac{RC_w}{R_w C'}}$$

~60-80 ps/mm

in 180 nm process